

Cluster Created

myCluster Updated 5 minutes ago [Terminate](#) [Clone in AWS CLI](#) [Clone](#)

Summary

Cluster info	Applications	Cluster management	Status and time
Cluster ID j-ZKLXL0CKNZQ	Amazon EMR version emr-5.20.1	Log destination in Amazon S3 aws-logs-767398109550-us-east-1/elasticmapreduce	Status Waiting
Cluster ARN arn:aws:elasticmapreduce:us-east-1:767398109550:cluster/j-ZKLXL0CKNZQ	Installed applications Ganglia 3.7.2, Hadoop 2.8.5, Hive 2.3.4, Hue 4.3.0, JupyterHub 0.9.4, Spark 2.4.0, Tez 0.9.1, Zeppelin 0.8.0	Primary node public DNS ec2-54-91-101-126.compute-1.amazonaws.com Connect to the Primary node using SSH Connect to the Primary node using SSM	Creation time 9 May 2026 11:00 (UTC+05:30)
Cluster configuration Instance groups			Elapsed time 21 minutes, 38 seconds
Capacity 1 Primary 1 Core 0 Task			

[Properties](#) [Bootstrap actions](#) [Instances \(hardware\)](#) [Steps](#) [Applications](#) [Configurations](#) [Monitoring](#) [Events](#) [Tags \(0\)](#)

Properties [Bootstrap actions](#) [Instances \(hardware\)](#) [Steps](#) [Applications](#) [Configurations](#) [Monitoring](#) [Events](#) [Tags \(0\)](#)

Cluster logs [Info](#)

Archive log files to Amazon S3
Turned on

Amazon S3 location
s3://aws-logs-767398109550-us-east-1/elasticmapreduce/ [l](#)

Turn on encryption for logs
Turned off

Cluster termination and node replacement [Info](#) [Edit](#)

Termination option	Idle time
Manually terminate cluster	-
Termination protection	Off
Unhealthy node replacement	On

Network and security [Info](#)

Network	Security configuration	Permissions
Virtual Private Cloud (VPC) vpc-0d6fa1f19bc2325eb l	Security configuration none	Service role for Amazon EMR EMR_DefaultRole l
Subnet ID and Availability Zone (AZ) subnet-004c08bcf08790615 l us-east-1c	EC2 key pair my-key-2	EC2 Instance profile EMR_EC2_DefaultRole
EC2 security groups (firewall)		Auto-scaling role Not configured
Primary node EMR-managed security group sg-05fd73a3e97dcfb45 l		
Additional security groups		

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us-east-1.console.aws.amazon.com/emr/home?region=us-east-1#/clusterDetails/f-ZKJLXLOCKNZQ

Account ID: 7673-9810-9550
vociabts/user5000584=roshanjadhaoo15@gmail.com

Amazon EMR > EMR on EC2: Clusters > myCluster

Applications installed on your Amazon EMR cluster publish user interfaces (UI) as websites. You can use these to monitor cluster activity.

On-cluster application UIs
On-cluster UIs are only available while your cluster is running. Use the following links to get started. To access all the application UIs, set up SSH tunnelling.

Application UIs on the primary node
These require SSH tunnelling to be enabled. [Enable an SSH connection](#)

Application	UI URL
Ganglia	http://ec2-54-91-101-126.compute-1.amazonaws.com/ganglia/
HDFS name node	http://ec2-54-91-101-126.compute-1.amazonaws.com:50070/
Hue	http://ec2-54-91-101-126.compute-1.amazonaws.com:8888/
JupyterHub	https://ec2-54-91-101-126.compute-1.amazonaws.com:9443/
Resource manager	http://ec2-54-91-101-126.compute-1.amazonaws.com:8088/
Spark history server	http://ec2-54-91-101-126.compute-1.amazonaws.com:18080/
Tez UI	http://ec2-54-91-101-126.compute-1.amazonaws.com:8080/tez-ui
Zeppelin	http://ec2-54-91-101-126.compute-1.amazonaws.com:8890/

Application UIs on the core and task nodes

Application	UI URL
HDFS data node	http://ec2-98-84-132-244.compute-1.amazonaws.com:50075/
Node manager	http://ec2-98-84-132-244.compute-1.amazonaws.com:8042/

ec2-54-91-101-126.compute-1.amazonaws.com:8890

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us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#SecurityGroup:group-id=sg-05fd73a3e97dcfb45

Account ID: 7673-9810-9550
vociabts/user5000584=roshanjadhaoo15@gmail.com

EC2 > Security Groups > sg-05fd73a3e97dcfb45 - ElasticMapReduce-master

Owner: 767398109550 Inbound rules count: 8 Permission entries Outbound rules count: 1 Permission entry

[Inbound rules](#) [Outbound rules](#) [Sharing](#) [VPC associations](#) [Related resources - new](#) [Tags](#)

Inbound rules (8) [Manage tags](#) [Edit inbound rules](#)

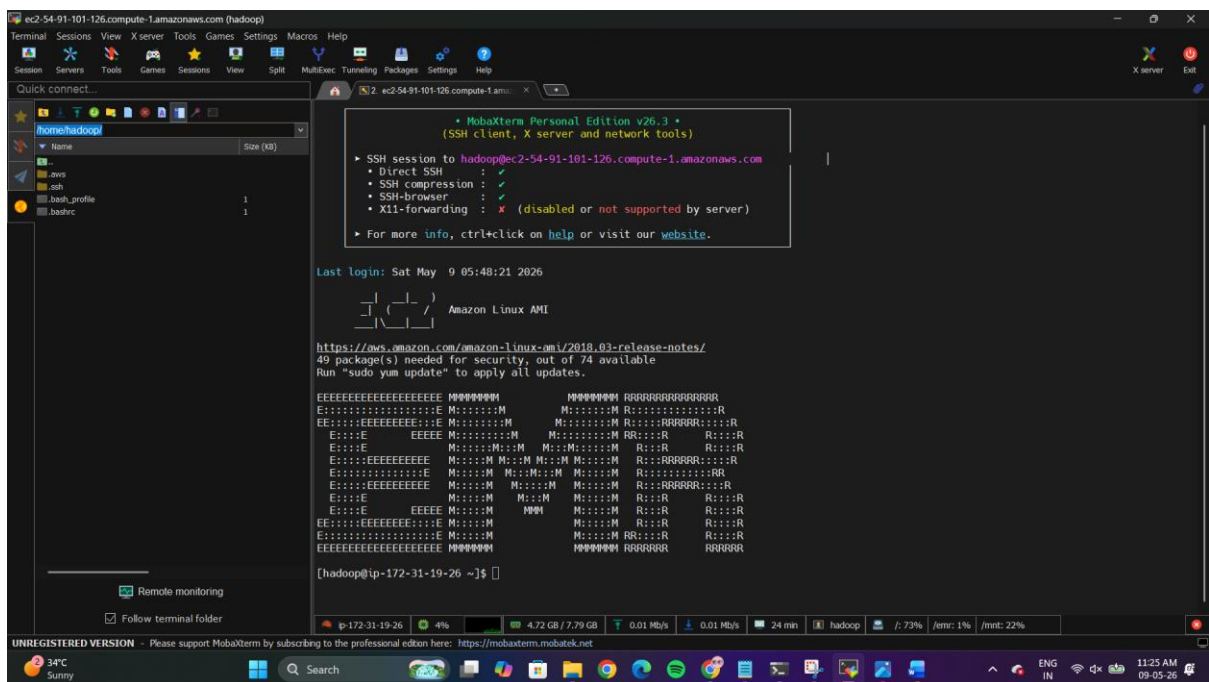
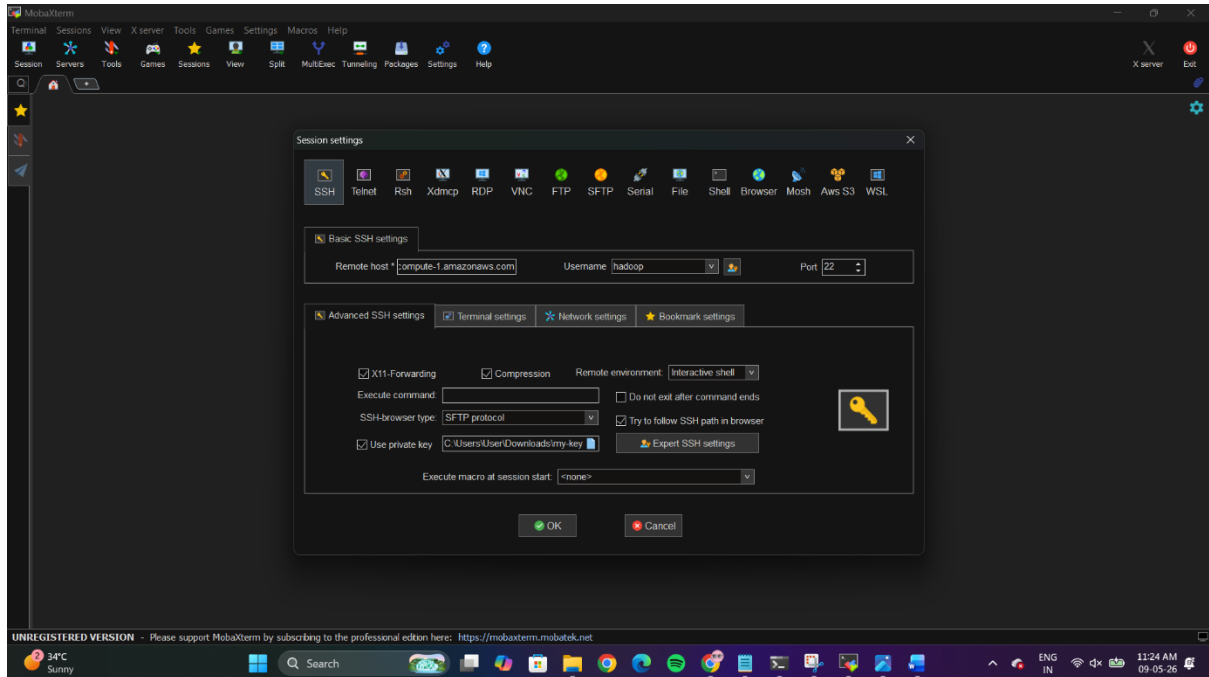
	Name	Security group rule ID	IP version	Type	Protocol	Port range
☐	-	sg-r-0933d43df45481c13	-	Custom TCP	TCP	8443
☐	-	sg-r-004f92f3ef544be9	-	All UDP	UDP	0 - 65535
☐	-	sg-r-05dcad99037473b6a	-	All ICMP - IPv4	ICMP	All
☐	-	sg-r-074ab51f850356949	-	All TCP	TCP	0 - 65535
☐	-	sg-r-07f3a6870ef039d03	IPv4	All traffic	All	All
☐	-	sg-r-0f9dc2c01b520465c	-	All UDP	UDP	0 - 65535
☐	-	sg-r-08d6333d85ac01fc6	-	All TCP	TCP	0 - 65535
☐	-	sg-r-0b51f3115f5be03e5	-	All ICMP - IPv4	ICMP	All

CloudShell Feedback

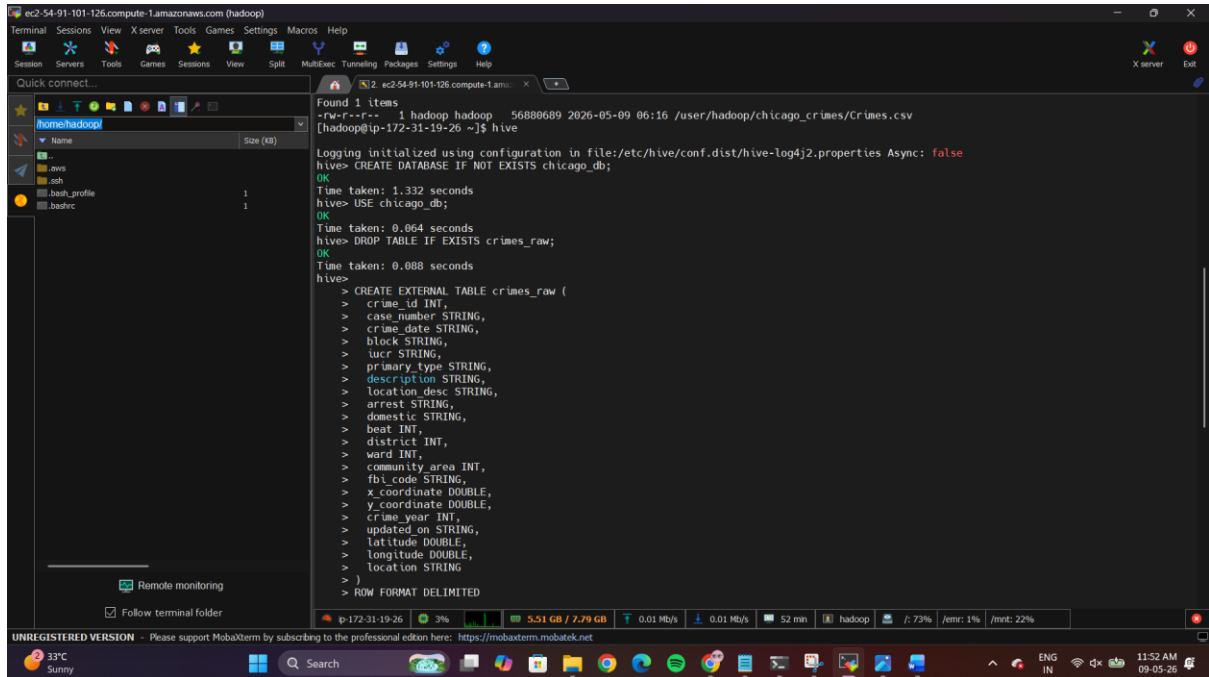
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Connecting MobaXterm With Remote Host, username and Private key



Creating External Table

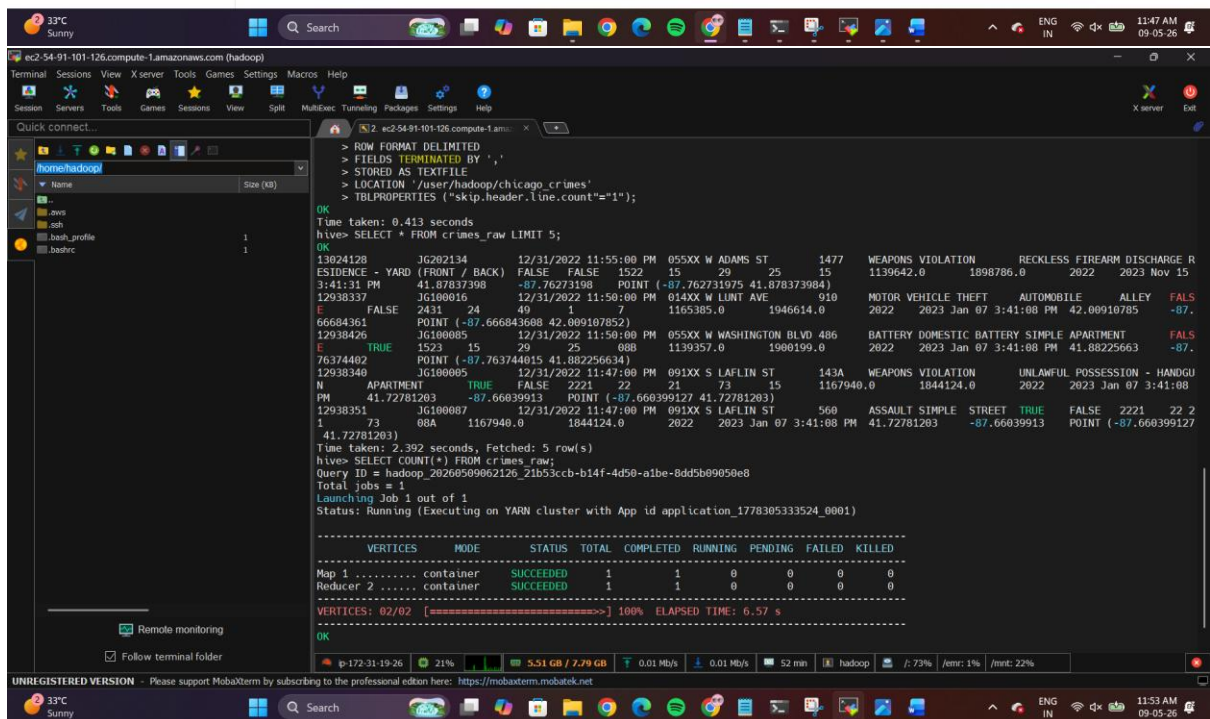
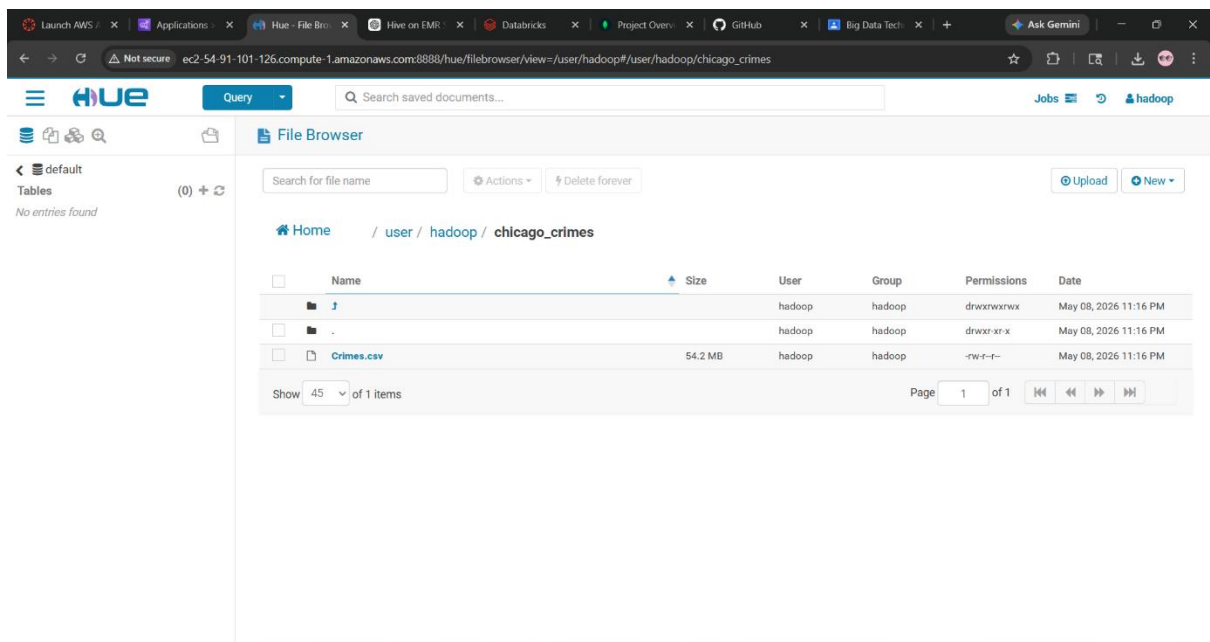


The screenshot shows a MobaXterm terminal window with a dark theme. The title bar indicates the connection is to 'ec2-54-91-101-126.compute-1.amazonaws.com (hadoop)'. The left sidebar shows a 'Quick connect...' section with a list of servers, including 'homehadoop' which is selected. The main terminal area displays the following commands and output:

```
Found 1 items
-rw-r--r-- 1 hadoop hadoop 56880689 2026-05-09 06:16 /user/hadoop/chicago_crimes/Crimes.csv
[hadoop@ip-172-31-19-26 ~]$ hive

Logging initialized using configuration in file:/etc/hive/conf.dist/hive-log4j2.properties Async: false
hive> CREATE DATABASE IF NOT EXISTS chicago_db;
OK
Time taken: 1.332 seconds
hive> USE chicago_db;
OK
Time taken: 0.064 seconds
hive> DROP TABLE IF EXISTS crimes_raw;
OK
Time taken: 0.088 seconds
hive>
> CREATE EXTERNAL TABLE crimes_raw (
>   crime_id INT,
>   case_number STRING,
>   crime_date STRING,
>   block STRING,
>   iucr STRING,
>   primary_type STRING,
>   description STRING,
>   location_desc STRING,
>   arrest STRING,
>   domestic STRING,
>   beat INT,
>   district INT,
>   ward INT,
>   community_area INT,
>   fb_code STRING,
>   x_coordinate DOUBLE,
>   y_coordinate DOUBLE,
>   crime_year INT,
>   updated_on STRING,
>   latitude DOUBLE,
>   longitude DOUBLE,
>   location STRING
> )
> ROW FORMAT DELIMITED
```

The bottom status bar of the terminal shows system information: 'UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: <https://mobaxterm.mobatek.net>'. The Windows taskbar at the very bottom shows the date and time as '11:52 AM 09-05-26'.



```

hive> DROP TABLE IF EXISTS crimes_partitioned;
OK
Time taken: 0.041 seconds
hive>
> CREATE TABLE crimes_partitioned (
>   crime_id INT,
>   case_number STRING,
>   crime_date STRING,
>   block STRING,
>   primary_type STRING,
>   description STRING,
>   location_desc STRING,
>   arrest STRING,
>   domestic STRING,
>   beat INT,
>   district INT,
>   ward INT,
>   community_area INT,
>   latitude DOUBLE,
>   longitude DOUBLE
> )
> PARTITIONED BY (crime_year INT)
> CLUSTERED BY (district) INTO 4 BUCKETS
> ROW FORMAT DELIMITED
> FIELDS TERMINATED BY ','
> STORED AS TEXTFILE;
OK
Time taken: 0.144 seconds

```

The screenshot shows a MobaXterm terminal window with a Hive session. The user has executed several commands to set Hive configuration properties and insert data into a partitioned table. The terminal output shows the execution of a SELECT query, followed by a summary of the query execution, including the number of vertices, modes, and status.

```

hive> SET hive.exec.dynamic.partition.mode=nonstrict;
hive> SET hive.exec.dynamic.partitions=10000;
hive> SET hive.exec.max.dynamic.partitions.pernode=10000;
hive> SET hive.enforce.bucketing=true;
hive> INSERT OVERWRITE TABLE crimes_partitioned
> PARTITION (crime_year=2022)
> SELECT
>   crime_id,
>   case_number,
>   crime_date,
>   block,
>   primary_type,
>   description,
>   location_desc,
>   arrest,
>   domestic,
>   beat,
>   district,
>   ward,
>   community_area,
>   latitude,
>   longitude
> FROM crimes_row
> WHERE crime_year = 2022;
Query ID = hadoop_20260509063129_e2bb434e-b0b8-4903-9e61-adb7b6930e93
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1778305333524_0002)

```

VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	container	SUCCEEDED	1	1	0	0	0	0
Reducer 2	container	SUCCEEDED	4	4	0	0	0	0

```

VERTICES: 02/02 [=====] 100% ELAPSED TIME: 15.88 s
Loading data to table chicago_db.crimes_partitioned partition (crime_year=2022)
OK
Time taken: 22.83 seconds

```

Query 1

```

hive> SELECT COUNT(*) AS total_crimes
> FROM crimes_partitioned;
OK
236756
Time taken: 0.187 seconds, Fetched: 1 row(s)
hive>

```

Query 2 – 3

```
hive> SELECT primary_type, COUNT(*) AS crime_count
> FROM crimes_partitioned
> GROUP BY primary_type
> ORDER BY crime_count DESC
> LIMIT 5;
Query ID = hadoop_20260509064313_495e9e96-08e8-42d2-a3f3-5be2551ba3d5
Total jobs = 1
Launching Job 1 out of 1
Tez session was closed. Reopening...
Session re-established.
Status: Running (Executing on YARN cluster with App id application_1778305333524_0003)
```

	VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1		container	SUCCEEDED	1	1	0	0	0	0
Reducer 2		container	SUCCEEDED	1	1	0	0	0	0
Reducer 3		container	SUCCEEDED	1	1	0	0	0	0

VERTICES: 03/03 [=====>] 100% ELAPSED TIME: 8.26 s

```
OK
THEFT 54883
BATTERY 38856
CRIMINAL DAMAGE 27248
MOTOR VEHICLE THEFT 20974
ASSAULT 20485
Time taken: 17.24 seconds, Fetched: 5 row(s)
hive> SELECT crime_year, COUNT(*) AS total_crimes
> FROM crimes_partitioned
> GROUP BY crime_year
> ORDER BY crime_year;
Query ID = hadoop_20260509064338_9bbfe116-d0eb-4fa5-b1da-5d2504ec1f45
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1778305333524_0003)
```

	VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1		container	SUCCEEDED	1	1	0	0	0	0
Reducer 2		container	SUCCEEDED	1	1	0	0	0	0
Reducer 3		container	SUCCEEDED	1	1	0	0	0	0

VERTICES: 03/03 [=====>] 100% ELAPSED TIME: 6.61 s

```
OK
2022 236756
Time taken: 7.476 seconds, Fetched: 1 row(s)
hive>
```

Query 4


```

Time taken: 7.478 seconds, Fetched: 1 row(s)
hive> SELECT primary_type, COUNT(*) AS arrest_count
> FROM crimes_partitioned
> WHERE LOWER(arrest)='true'
> GROUP BY primary_type
> HAVING COUNT(*) > 500
> ORDER BY arrest_count DESC;
Query ID = hadoop_20260509064455_bbaff7e5-2dc9-47b3-945b-17af1750848a
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1778305333524_0003)

```

	VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1	container		SUCCEEDED	1	1	0	0	0	0
Reducer 2	container		SUCCEEDED	1	1	0	0	0	0
Reducer 3	container		SUCCEEDED	1	1	0	0	0	0

VERTICES: 03/03 [=====] 100% ELAPSED TIME: 7.46 s

```

OK
WEAPONS VIOLATION      5394
BATTERY 5173
NARCOTICS              4645
OTHER OFFENSE         2094
THEFT 2029
ASSAULT 1900
CRIMINAL TRESPASS     1246
CRIMINAL DAMAGE 899
MOTOR VEHICLE THEFT   583
ROBBERY 558
Time taken: 8.244 seconds, Fetched: 10 row(s)
hive> █

```

Query 5

```

hive> EXPLAIN
> SELECT *
> FROM crimes_partitioned
> WHERE crime_year=2022;
OK
Plan optimized by CBO.

Stage-0
Fetch Operator
  limit:-1
  Select Operator [SEL_2]
    Output:["_col0","_col1","_col2","_col3","_col4","_col5","_col6","_col7","_col8","_col9","_col10","_col11","_col12","_col13","_col14","_col15"]
    TableScan [TS_0]
      Output:["crime_id","case_number","crime_date","block","primary_type","description","location_desc","arrest","domestic","beat","district","ward","community_area","latitude","longitude"]

Time taken: 0.171 seconds, Fetched: 10 row(s)
hive> █

```

```

Time taken: 0.171 seconds, Fetched: 10 row(s)
hive> SELECT *
> FROM crimes_partitioned
> WHERE crime_year=2022
> LIMIT 20;
OK
12582064 JF100392 01/01/2022 12:00:00 AM 019XX S HALSTED ST THEFT $500 AND UNDER VEHICLE NON-COMMERCIAL FALSE FALS
E 1235 12 11 31 41.85637974 -87.64659823 2022
13435498 JH232904 01/01/2022 12:00:00 AM 056XX N CENTRAL AVE BATTERY DOMESTIC BATTERY SIMPLE APARTMENT FALSE TRUE
1622 16 45 11 NULL NULL 2022
12677514 JF216088 01/01/2022 12:00:00 AM 054XX N KENMORE AVE CRIMINAL DAMAGE TO PROPERTY OTHER (SPECIFY) FALSE FALS
E 2023 20 48 77 41.98050242 -87.65652771 2022
13051561 JG234761 01/01/2022 12:00:00 AM 017XX W 16TH ST WEAPONS VIOLATION RECKLESS FIREARM DISCHARGE STREET FALS
E FALSE 1234 12 28 28 NULL NULL 2022
14175370 JK226975 01/01/2022 12:00:00 AM 041XX W 69TH ST BATTERY SIMPLE RESIDENCE FALSE FALSE 833 8 23 6
5 NULL NULL 2022
13126333 JG323739 01/01/2022 12:00:00 AM 019XX W MAYPOLE AVE OFFENSE INVOLVING CHILDREN CRIMINAL SEXUAL ABUSE BY FAM
ILY MEMBER APARTMENT TRUE TRUE 1223 12 27 28 NULL NULL 2022
12581817 JF100052 01/01/2022 12:00:00 AM 080XX S CRANDON AVE BATTERY DOMESTIC BATTERY SIMPLE RESIDENCE TRUE TRUE
414 4 8 46 41.749261 -87.56862932 2022
12582422 JF100497 01/01/2022 12:00:00 AM 060XX W MELROSE ST CRIMINAL DAMAGE TO VEHICLE STREET FALSE FALSE 1633
16 30 17 41.9393487 -87.77737954 2022
12691932 JF233522 01/01/2022 12:00:00 AM 024XX E 106TH ST MOTOR VEHICLE THEFT AUTOMOBILE STREET FALSE TRUE
434 4 7 51 41.70276162 -87.56373152 2022
12591964 JF112288 01/01/2022 12:00:00 AM 001XX S ASHLAND AVE THEFT OVER $500 CONSTRUCTION SITE FALSE FALS
E 1224 12 28 28 41.87964349 -87.66675118 2022
12607743 JF131728 01/01/2022 12:00:00 AM 068XX W HOBART AVE DECEPTIVE PRACTICE FINANCIAL IDENTITY THEFT $300 AND UN
DER RESIDENCE FALSE FALSE 1612 16 41 10 41.98374633 -87.7997238 2022
12613727 JF139205 01/01/2022 12:00:00 AM 034XX W 84TH ST THEFT OVER $500 RESIDENCE FALSE FALSE 834 8 1
8 70 41.74057926 -87.70762474 2022
14165892 JK216512 01/01/2022 12:00:00 AM 017XX W MORSE AVE DECEPTIVE PRACTICE FRAUD OR CONFIDENCE GAME APAR
TMENT FALSE FALSE 2424 24 49 1 NULL NULL 2022
12797476 JF360732 01/01/2022 12:00:00 AM 040XX E 134TH ST OFFENSE INVOLVING CHILDREN AGGRAVATED CRIMINAL SEXUAL A
BUSE BY FAMILY MEMBER RESIDENCE FALSE TRUE 433 4 10 55 41.65143133 -87.52725823 2022
14126587 JK168718 01/01/2022 12:00:00 AM 019XX W HURON ST OFFENSE INVOLVING CHILDREN AGGRAVATED CRIMINAL SEXUAL A
BUSE BY FAMILY MEMBER RESIDENCE FALSE TRUE 1221 12 36 24 NULL NULL 2022
12582722 JF101055 01/01/2022 12:00:00 AM 082XX S MARQUETTE AVE OTHER OFFENSE HARASSMENT BY TELEPHONE RESIDENCE FALS
E TRUE 423 4 7 46 41.74568005 -87.5588512 2022
12985196 JG155434 01/01/2022 12:00:00 AM 032XX N NARRAGANSETT AVE OTHER OFFENSE HARASSMENT BY TELEPHONE RESIDENCE F

```


Query6

```
hive> SELECT district, COUNT(*) AS domestic_crime_count
> FROM crimes_partitioned
> WHERE LOWER(domestic)='true'
> GROUP BY district
> ORDER BY domestic_crime_count DESC
> LIMIT 3;
Query ID = hadoop_20260509064715_44605402-a412-4a01-a820-2a9a1d0d0a11
Total jobs = 1
Launching Job 1 out of 1
Status: Running (Executing on YARN cluster with App id application_1778305333524_0003)
```

	VERTICES	MODE	STATUS	TOTAL	COMPLETED	RUNNING	PENDING	FAILED	KILLED
Map 1		container	SUCCEEDED	1	1	0	0	0	0
Reducer 2		container	SUCCEEDED	1	1	0	0	0	0
Reducer 3		container	SUCCEEDED	1	1	0	0	0	0

VERTICES: 03/03 [=====>>>] 100% ELAPSED TIME: 7.91 s

OK

6 3821

4 3448

8 3048

Time taken: 8.624 seconds, Fetched: 3 row(s)

hive>